

In one line: The **Keychain** is an encrypted database *outside* your sandbox (securityd); its rows open with **class keys** that depend on lock state. The **Secure Enclave** is a separate chip: keys *born* there **never leave** — data in, signature out. Biometrics are secure only when they **gate a key**, never when they return a Bool.

Keychain

- SecItemAdd · SecItemCopyMatching · SecItemUpdate · SecItemDelete — dictionary in, OSStatus out. **No upsert:** add, and on errSecDuplicateItem update; absent → errSecItemNotFound.
- kSecClass: GenericPassword (tokens; identity = service + account), InternetPassword, Key, Certificate, Identity. Reads need kSecReturnData: true + kSecMatchLimit.
- **Sharing:** kSecAttrAccessGroup, same Team ID, group in the Keychain Sharing entitlement. kSecAttrSynchronizable = iCloud Keychain. Items usually **survive uninstall** — not a contract; never a “first run” flag.

kSecAttrAccessible... — choose by *when you read*

WhenPasscodeSet-ThisDeviceOnly	unlocked + passcode; deleted if the passcode is removed
WhenUnlocked (default)	only while unlocked — UI-driven secrets
AfterFirstUnlock	1st unlock → reboot, also locked: background work
...ThisDeviceOnly	never migrates (no iCloud sync, no restore to a new device)

Secure Enclave & biometrics

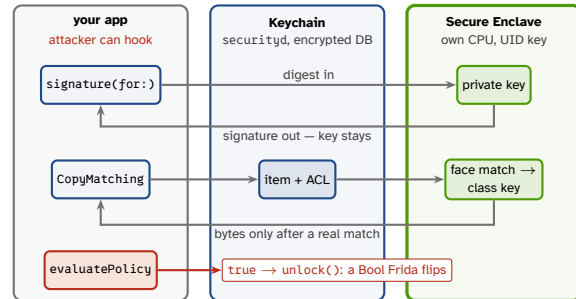
- SE keys: **EC P-256** (signing + ECDH); since iOS 26 CryptoKit adds **ML-KEM-768/1024** and **ML-DSA-65/87**. No RSA, no arbitrary blobs: generated and used *inside*; the app holds an opaque reference. CryptoKit SecureEnclave.P256.Signing.PrivateKey — its dataRepresentation is a wrapped handle only *this* SE can use (store it in the Keychain). No SE on the Simulator.
- Protect a secret *with* the SE: encrypt it under an SE-bound key, or gate its item with a SecAccessControl.
- Flags: .biometryCurrentSet (item dies when a face/finger is enrolled — an attacker adding theirs loses access), .biometryAny, .userPresence (biometry *or* passcode), .privateKeyUsage (SE keys).
- LAContext.evaluatePolicy = a Bool in hookable code: UX only. Reuse it via kSecUseAuthenticationContext (one prompt). Face ID needs NSFaceIDUsageDescription.

Example

```
let id: [CFString: Any] = [kSecClass: kSecClassGenericPassword,
  kSecAttrService: "api", kSecAttrAccount: "token"]
var add = id; add[kSecValueData] = data
add[kSecAttrAccessible] = kSecAttrAccessibleAfterFirstUnlock
var st = SecItemAdd(add as CFDictionary, nil)
if st == errSecDuplicateItem { // no upsert
  let new = [kSecValueData: data] as CFDictionary
  st = SecItemUpdate(id as CFDictionary, new) }
let acl = SecAccessControlCreateWithFlags(nil,
  kSecAttrAccessibleWhenUnlockedThisDeviceOnly,
  [.privateKeyUsage, .biometryCurrentSet], nil)!
let key = try SecureEnclave.P256.Signing
  .PrivateKey(accessControl: acl)
let sig = try key.signature(for: nonce) // Face ID, signed in SE
```

Related: app hardening & privacy · CryptoKit (AES · GCM, HKDF) · App Attest (SE-backed) · passkeys · networking / pinning

The trust boundary



Data Protection — files, same idea

.complete	unlocked only; key dropped 10 s after lock (Require Password: Immediately)
.completeUnlessOpen	open files keep working when locked; new files can be created
.completeUntilFirstUserAuthentication	default — the twin of AfterFirstUnlock
.none	UID-key only, always readable

data.write(to:options: .completeFileProtection);
 SQLite: the -wal/-shm files too. **No passcode** ⇒ **no passcode-derived keys**: every class degrades to .none (*unverified*).

Traps

- ! Background upload fails at 3 am → token is WhenUnlocked; locked read = errSecInteractionNotAllowed. Use AfterFirstUnlock.
- ! “Put the JWT in the Secure Enclave” — impossible; it holds keys.
- ! if faceIDOK { showVault() } is a UI gate, not security.
- ! .complete Core Data store + background launch = crash.
- ! Tokens in UserDefaults — a plain plist in the container.
- ! Add succeeds, read says errSecItemNotFound → query differs (service/account/access group) or no kSecReturnData.
- ! ...ThisDeviceOnly + kSecAttrSynchronizable — contradictory; synced items cannot be device-only.

Remember

“Keychain for secrets, class for *when*, enclave for keys, biometry on the key — never on a Bool.”